Configuração 10: M = 256, C = 8, K = 8, IoT = 20 dB, 16-QAM

Perda de SNR do BCD-MMSE (iter 4) em SER = 10^{-2}, em dB, em relação à referência ideal (float64); × = não chega a 10^{-2} nas SNRs simuladas

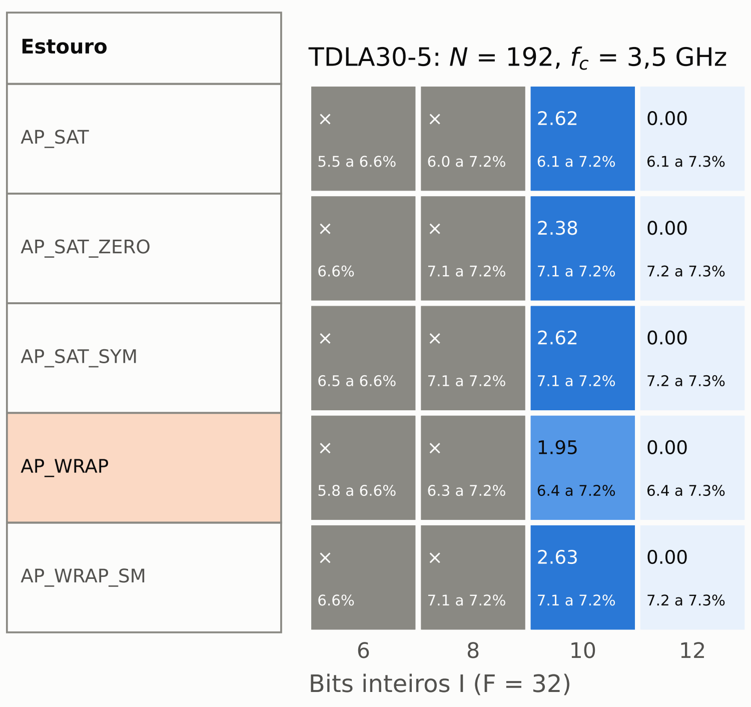
Fundo laranja no rótulo: o modo com o melhor resultado (estouro: pela varredura de I; quantização: pela varredura de F), comparando de menos bits para mais, sem contar a coluna em que tudo é zero

Faixa "x a y": o outro modo muda o valor (a quantização no mapa de I, o estouro no mapa de F); se os dois extremos têm cores diferentes, a célula é dividida na diagonal (em cima, a cor do menor; embaixo, a do maior)

Embaixo da perda: maior ocupação da XCU50 entre LUT, DSP, FF e BRAM (síntese HLS, sem diretivas), no mesmo formato e modo; sem número onde o formato e o modo não foram sintetizados

Perda de SNR (dB)

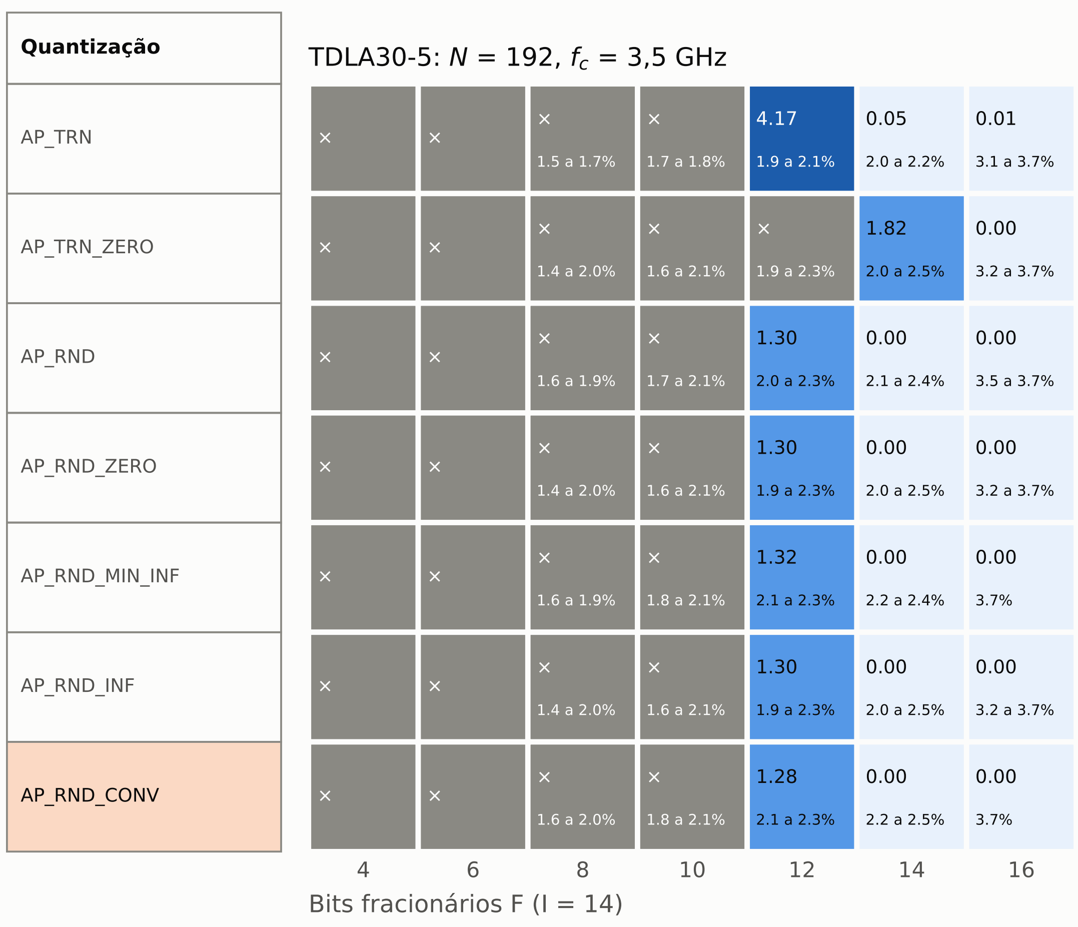
▢ < 0,1 dB
■ 0,1 a 0,5 dB
■ 0,5 a 1 dB
■ 1 a 2 dB
■ 2 a 3 dB
■ 3 a 5 dB
■ 5 a 10 dB
■ > 10 dB
■ × não chega a 10^{-2}
■ rótulo do melhor modo



TDLA30-5: N = 192, f_c = 3,5 GHz

Estouro	6	8	10	12
AP_SAT	× 5.5 a 6.6%	× 6.0 a 7.2%	2.62 6.1 a 7.2%	0.00 6.1 a 7.3%
AP_SAT_ZERO	× 6.6%	× 7.1 a 7.2%	2.38 7.1 a 7.2%	0.00 7.2 a 7.3%
AP_SAT_SYM	× 6.5 a 6.6%	× 7.1 a 7.2%	2.62 7.1 a 7.2%	0.00 7.2 a 7.3%
AP_WRAP	× 5.8 a 6.6%	× 6.3 a 7.2%	1.95 6.4 a 7.2%	0.00 6.4 a 7.3%
AP_WRAP_SM	× 6.6%	× 7.1 a 7.2%	2.63 7.1 a 7.2%	0.00 7.2 a 7.3%

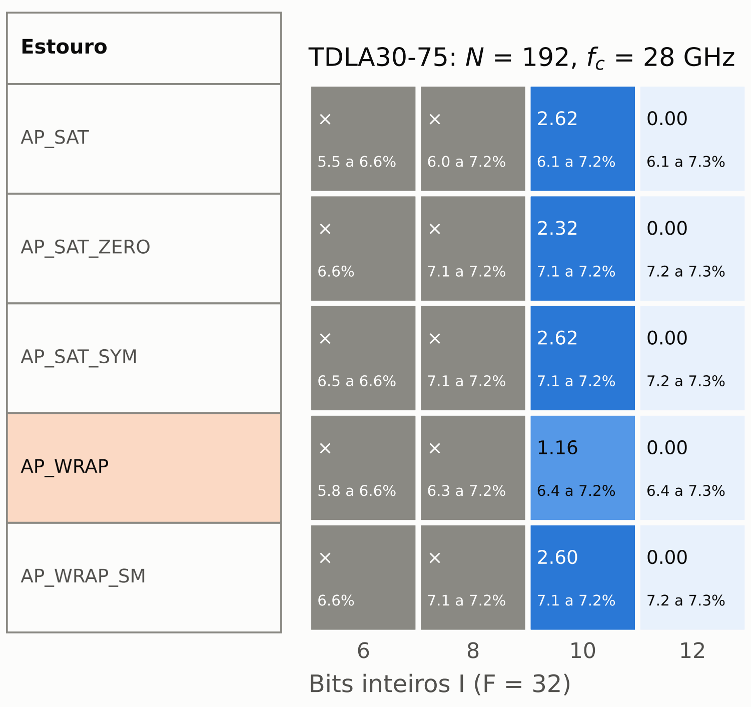
Bits inteiros I (F = 32)



TDLA30-5: N = 192, f_c = 3,5 GHz

Quantização	4	6	8	10	12	14	16
AP_TRN	×	×	× 1.5 a 1.7%	× 1.7 a 1.8%	4.17 1.9 a 2.1%	0.05 2.0 a 2.2%	0.01 3.1 a 3.7%
AP_TRN_ZERO	×	×	× 1.4 a 2.0%	× 1.6 a 2.1%	× 1.9 a 2.3%	1.82 2.0 a 2.5%	0.00 3.2 a 3.7%
AP_RND	×	×	× 1.6 a 1.9%	× 1.7 a 2.1%	1.30 2.0 a 2.3%	0.00 2.1 a 2.4%	0.00 3.5 a 3.7%
AP_RND_ZERO	×	×	× 1.4 a 2.0%	× 1.6 a 2.1%	1.30 1.9 a 2.3%	0.00 2.0 a 2.5%	0.00 3.2 a 3.7%
AP_RND_MIN_INF	×	×	× 1.6 a 1.9%	× 1.8 a 2.1%	1.32 2.1 a 2.3%	0.00 2.2 a 2.4%	0.00 3.7%
AP_RND_INF	×	×	× 1.4 a 2.0%	× 1.6 a 2.1%	1.30 1.9 a 2.3%	0.00 2.0 a 2.5%	0.00 3.2 a 3.7%
AP_RND_CONV	×	×	× 1.6 a 2.0%	× 1.8 a 2.1%	1.28 2.1 a 2.3%	0.00 2.2 a 2.5%	0.00 3.7%

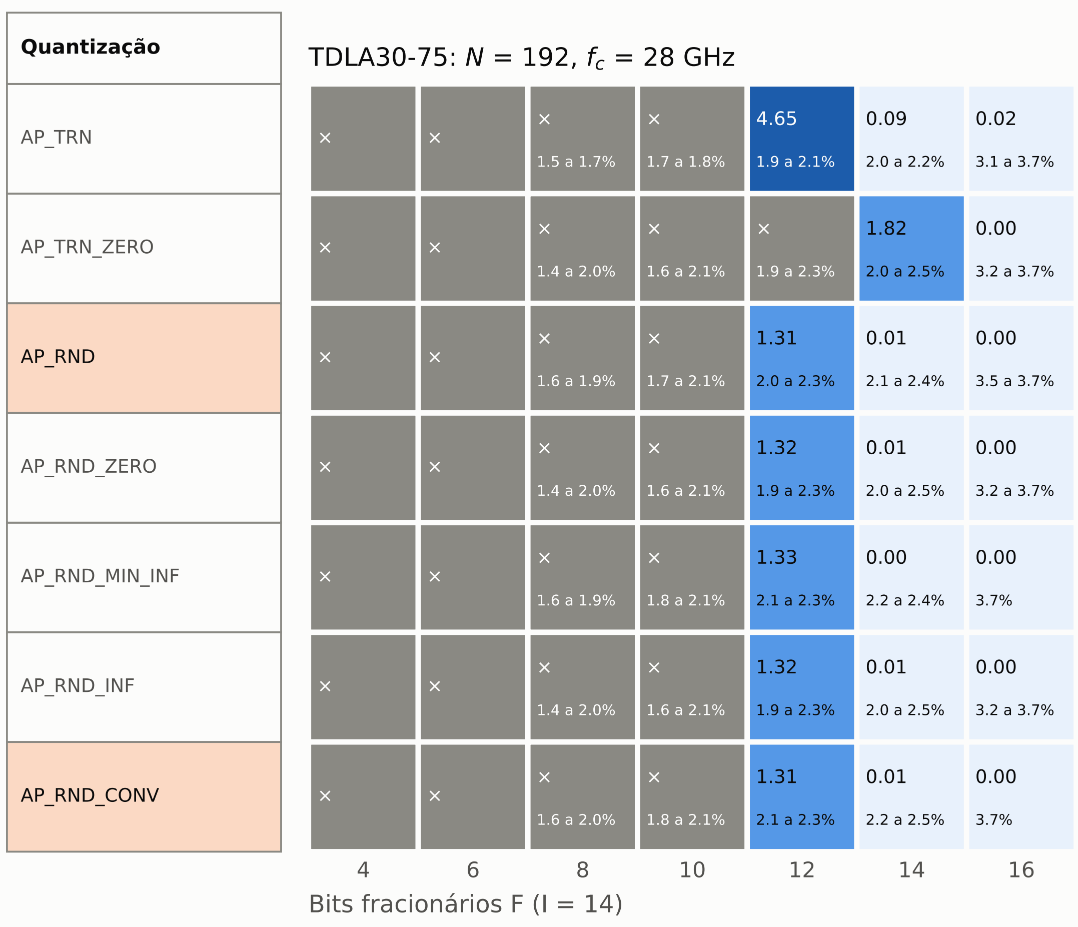
Bits fracionários F (I = 14)



TDLA30-75: N = 192, f_c = 28 GHz

Estouro	6	8	10	12
AP_SAT	× 5.5 a 6.6%	× 6.0 a 7.2%	2.62 6.1 a 7.2%	0.00 6.1 a 7.3%
AP_SAT_ZERO	× 6.6%	× 7.1 a 7.2%	2.32 7.1 a 7.2%	0.00 7.2 a 7.3%
AP_SAT_SYM	× 6.5 a 6.6%	× 7.1 a 7.2%	2.62 7.1 a 7.2%	0.00 7.2 a 7.3%
AP_WRAP	× 5.8 a 6.6%	× 6.3 a 7.2%	1.16 6.4 a 7.2%	0.00 6.4 a 7.3%
AP_WRAP_SM	× 6.6%	× 7.1 a 7.2%	2.60 7.1 a 7.2%	0.00 7.2 a 7.3%

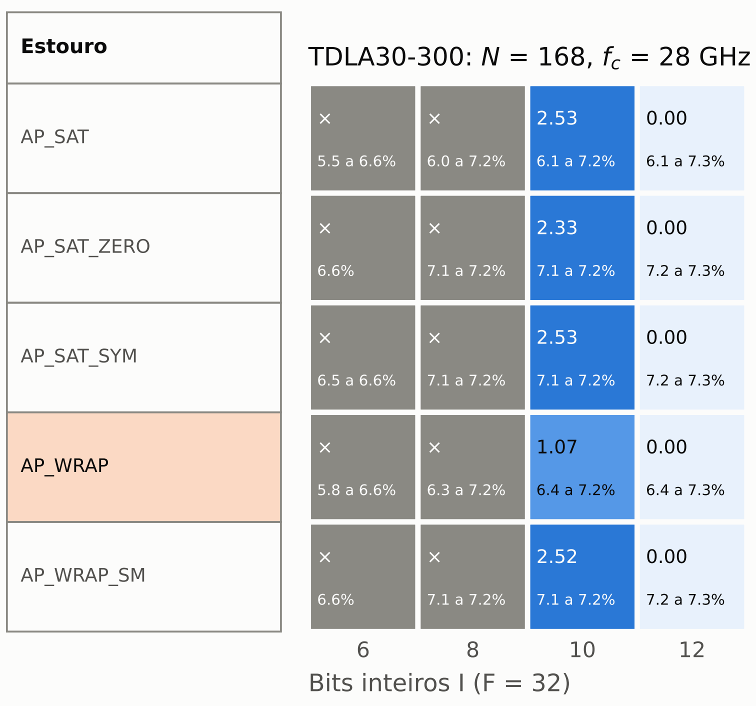
Bits inteiros I (F = 32)



TDLA30-75: N = 192, f_c = 28 GHz

Quantização	4	6	8	10	12	14	16
AP_TRN	×	×	× 1.5 a 1.7%	× 1.7 a 1.8%	4.65 1.9 a 2.1%	0.09 2.0 a 2.2%	0.02 3.1 a 3.7%
AP_TRN_ZERO	×	×	× 1.4 a 2.0%	× 1.6 a 2.1%	× 1.9 a 2.3%	1.82 2.0 a 2.5%	0.00 3.2 a 3.7%
AP_RND	×	×	× 1.6 a 1.9%	× 1.7 a 2.1%	1.31 2.0 a 2.3%	0.01 2.1 a 2.4%	0.00 3.5 a 3.7%
AP_RND_ZERO	×	×	× 1.4 a 2.0%	× 1.6 a 2.1%	1.32 1.9 a 2.3%	0.01 2.0 a 2.5%	0.00 3.2 a 3.7%
AP_RND_MIN_INF	×	×	× 1.6 a 1.9%	× 1.8 a 2.1%	1.33 2.1 a 2.3%	0.00 2.2 a 2.4%	0.00 3.7%
AP_RND_INF	×	×	× 1.4 a 2.0%	× 1.6 a 2.1%	1.32 1.9 a 2.3%	0.01 2.0 a 2.5%	0.00 3.2 a 3.7%
AP_RND_CONV	×	×	× 1.6 a 2.0%	× 1.8 a 2.1%	1.31 2.1 a 2.3%	0.01 2.2 a 2.5%	0.00 3.7%

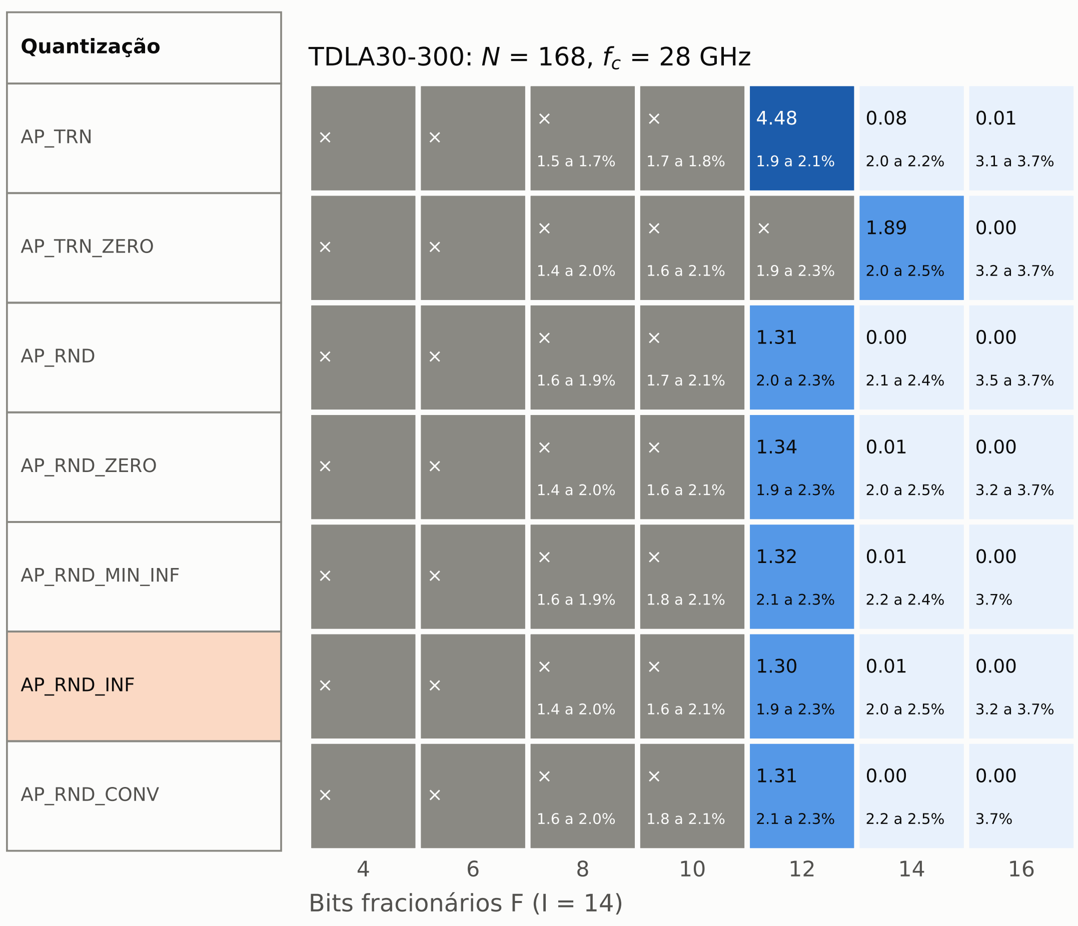
Bits fracionários F (I = 14)



TDLA30-300: N = 168, f_c = 28 GHz

Estouro	6	8	10	12
AP_SAT	× 5.5 a 6.6%	× 6.0 a 7.2%	2.53 6.1 a 7.2%	0.00 6.1 a 7.3%
AP_SAT_ZERO	× 6.6%	× 7.1 a 7.2%	2.33 7.1 a 7.2%	0.00 7.2 a 7.3%
AP_SAT_SYM	× 6.5 a 6.6%	× 7.1 a 7.2%	2.53 7.1 a 7.2%	0.00 7.2 a 7.3%
AP_WRAP	× 5.8 a 6.6%	× 6.3 a 7.2%	1.07 6.4 a 7.2%	0.00 6.4 a 7.3%
AP_WRAP_SM	× 6.6%	× 7.1 a 7.2%	2.52 7.1 a 7.2%	0.00 7.2 a 7.3%

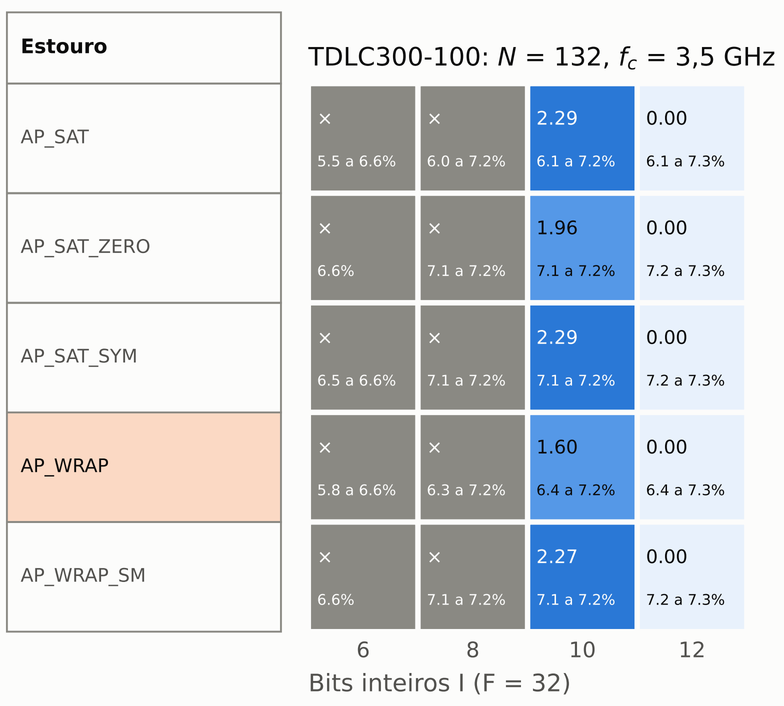
Bits inteiros I (F = 32)



TDLA30-300: N = 168, f_c = 28 GHz

Quantização	4	6	8	10	12	14	16
AP_TRN	×	×	× 1.5 a 1.7%	× 1.7 a 1.8%	4.48 1.9 a 2.1%	0.08 2.0 a 2.2%	0.01 3.1 a 3.7%
AP_TRN_ZERO	×	×	× 1.4 a 2.0%	× 1.6 a 2.1%	× 1.9 a 2.3%	1.89 2.0 a 2.5%	0.00 3.2 a 3.7%
AP_RND	×	×	× 1.6 a 1.9%	× 1.7 a 2.1%	1.31 2.0 a 2.3%	0.00 2.1 a 2.4%	0.00 3.5 a 3.7%
AP_RND_ZERO	×	×	× 1.4 a 2.0%	× 1.6 a 2.1%	1.34 1.9 a 2.3%	0.01 2.0 a 2.5%	0.00 3.2 a 3.7%
AP_RND_MIN_INF	×	×	× 1.6 a 1.9%	× 1.8 a 2.1%	1.32 2.1 a 2.3%	0.01 2.2 a 2.4%	0.00 3.7%
AP_RND_INF	×	×	× 1.4 a 2.0%	× 1.6 a 2.1%	1.30 1.9 a 2.3%	0.01 2.0 a 2.5%	0.00 3.2 a 3.7%
AP_RND_CONV	×	×	× 1.6 a 2.0%	× 1.8 a 2.1%	1.31 2.1 a 2.3%	0.00 2.2 a 2.5%	0.00 3.7%

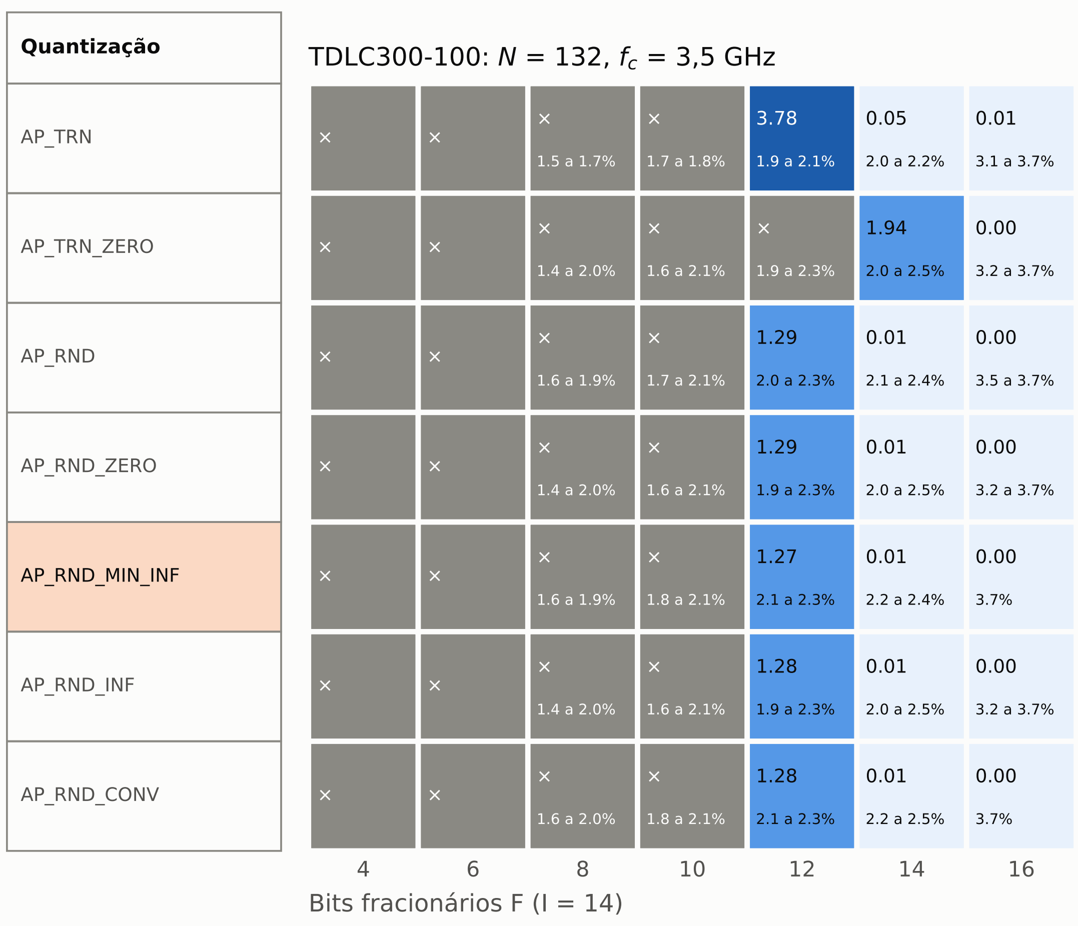
Bits fracionários F (I = 14)



TDLC300-100: N = 132, f_c = 3,5 GHz

Estouro	6	8	10	12
AP_SAT	× 5.5 a 6.6%	× 6.0 a 7.2%	2.29 6.1 a 7.2%	0.00 6.1 a 7.3%
AP_SAT_ZERO	× 6.6%	× 7.1 a 7.2%	1.96 7.1 a 7.2%	0.00 7.2 a 7.3%
AP_SAT_SYM	× 6.5 a 6.6%	× 7.1 a 7.2%	2.29 7.1 a 7.2%	0.00 7.2 a 7.3%
AP_WRAP	× 5.8 a 6.6%	× 6.3 a 7.2%	1.60 6.4 a 7.2%	0.00 6.4 a 7.3%
AP_WRAP_SM	× 6.6%	× 7.1 a 7.2%	2.27 7.1 a 7.2%	0.00 7.2 a 7.3%

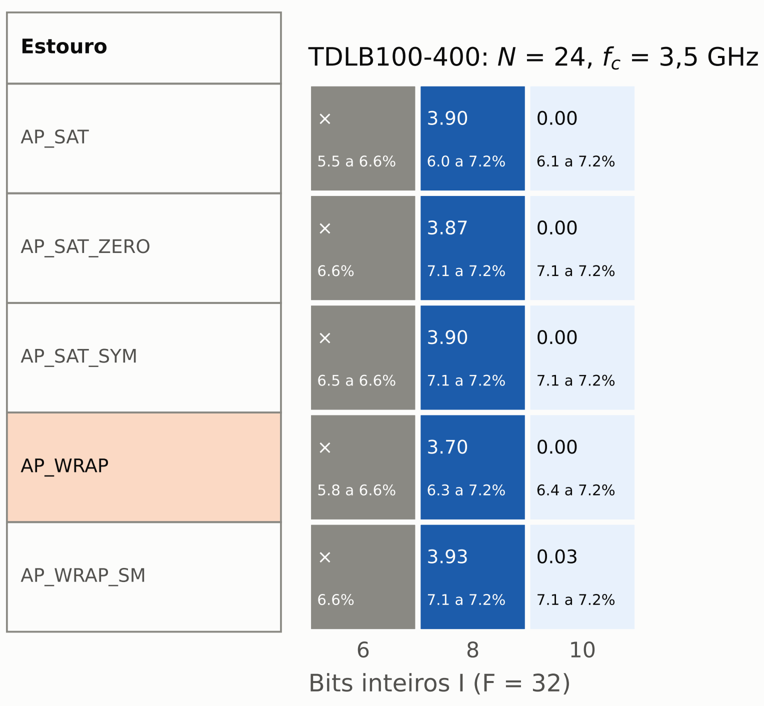
Bits inteiros I (F = 32)



TDLC300-100: N = 132, f_c = 3,5 GHz

Quantização	4	6	8	10	12	14	16
AP_TRN	×	×	× 1.5 a 1.7%	× 1.7 a 1.8%	3.78 1.9 a 2.1%	0.05 2.0 a 2.2%	0.01 3.1 a 3.7%
AP_TRN_ZERO	×	×	× 1.4 a 2.0%	× 1.6 a 2.1%	× 1.9 a 2.3%	1.94 2.0 a 2.5%	0.00 3.2 a 3.7%
AP_RND	×	×	× 1.6 a 1.9%	× 1.7 a 2.1%	1.29 2.0 a 2.3%	0.01 2.1 a 2.4%	0.00 3.5 a 3.7%
AP_RND_ZERO	×	×	× 1.4 a 2.0%	× 1.6 a 2.1%	1.29 1.9 a 2.3%	0.01 2.0 a 2.5%	0.00 3.2 a 3.7%
AP_RND_MIN_INF	×	×	× 1.6 a 1.9%	× 1.8 a 2.1%	1.27 2.1 a 2.3%	0.01 2.2 a 2.4%	0.00 3.7%
AP_RND_INF	×	×	× 1.4 a 2.0%	× 1.6 a 2.1%	1.28 1.9 a 2.3%	0.01 2.0 a 2.5%	0.00 3.2 a 3.7%
AP_RND_CONV	×	×	× 1.6 a 2.0%	× 1.8 a 2.1%	1.28 2.1 a 2.3%	0.01 2.2 a 2.5%	0.00 3.7%

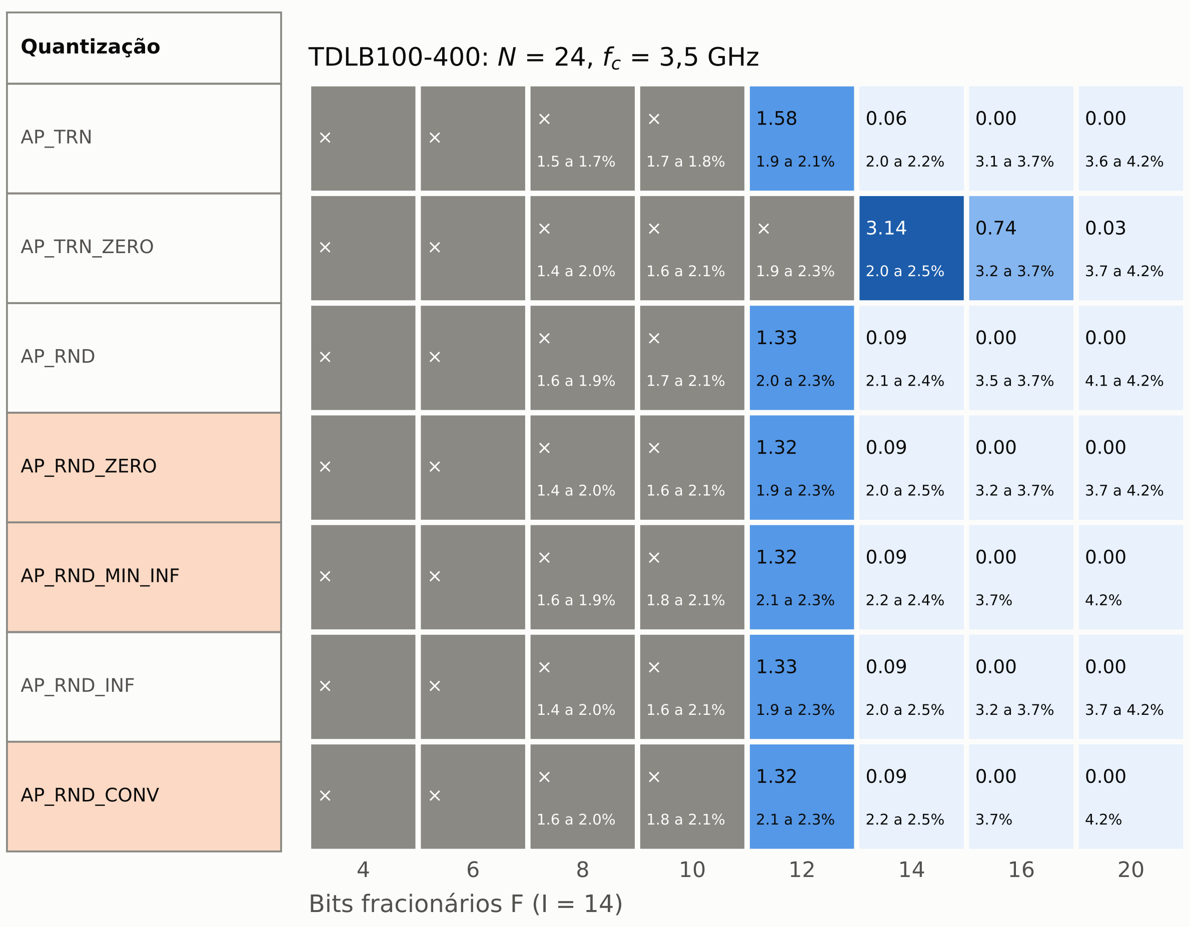
Bits fracionários F (I = 14)



TDLB100-400: N = 24, f_c = 3,5 GHz

Estouro	6	8	10
AP_SAT	× 5.5 a 6.6%	3.90 6.0 a 7.2%	0.00 6.1 a 7.2%
AP_SAT_ZERO	× 6.6%	3.87 7.1 a 7.2%	0.00 7.1 a 7.2%
AP_SAT_SYM	× 6.5 a 6.6%	3.90 7.1 a 7.2%	0.00 7.1 a 7.2%
AP_WRAP	× 5.8 a 6.6%	3.70 6.3 a 7.2%	0.00 6.4 a 7.2%
AP_WRAP_SM	× 6.6%	3.93 7.1 a 7.2%	0.03 7.1 a 7.2%

Bits inteiros I (F = 32)



TDLB100-400: N = 24, f_c = 3,5 GHz

Quantização	4	6	8	10	12	14	16	20
AP_TRN	×	×	× 1.5 a 1.7%	× 1.7 a 1.8%	1.58 1.9 a 2.1%	0.06 2.0 a 2.2%	0.00 3.1 a 3.7%	0.00 3.6 a 4.2%
AP_TRN_ZERO	×	×	× 1.4 a 2.0%	× 1.6 a 2.1%	× 1.9 a 2.3%	3.14 2.0 a 2.5%	0.74 3.2 a 3.7%	0.03 3.7 a 4.2%
AP_RND	×	×	× 1.6 a 1.9%	× 1.7 a 2.1%	1.33 2.0 a 2.3%	0.09 2.1 a 2.4%	0.00 3.5 a 3.7%	0.00 4.1 a 4.2%
AP_RND_ZERO	×	×	× 1.4 a 2.0%	× 1.6 a 2.1%	1.32 1.9 a 2.3%	0.09 2.0 a 2.5%	0.00 3.2 a 3.7%	0.00 3.7 a 4.2%
AP_RND_MIN_INF	×	×	× 1.6 a 1.9%	× 1.8 a 2.1%	1.32 2.1 a 2.3%	0.09 2.2 a 2.4%	0.00 3.7%	0.00 4.2%
AP_RND_INF	×	×	× 1.4 a 2.0%	× 1.6 a 2.1%	1.33 1.9 a 2.3%	0.09 2.0 a 2.5%	0.00 3.2 a 3.7%	0.00 3.7 a 4.2%
AP_RND_CONV	×	×	× 1.6 a 2.0%	× 1.8 a 2.1%	1.32 2.1 a 2.3%	0.09 2.2 a 2.5%	0.00 3.7%	0.00 4.2%

Bits fracionários F (I = 14)